Experiment :- study of LAN Topology Network device and Address Configuration

Objective :-

1) To study the LAN topology Used in College campus.

2) To identity networking device Understand their function

3) To Analyze wired and wireless communication method Used in the network.

4) To measure speed of network for wired and wireless connection.

5) To obtained MAC, IP Address & subnet mask.

* Problem statement - 1

- Draw & identify LAN topology of College Campus & clearly label all components such as router switches & etc.

Soln :- The College Campus network observed this study consists of :

* Layer-3 Core switches in the Server room.
* Managed by PoE + Switches
* Wireless Access point (APs)
* Ethernet port in classroom & laborations

→ Network Topology:

- Hybrid Hierarchical topology which combines:

(i) Tree topology
(ii) Extended star topology
(iii) Ring topology

→ The Network Hierarchical layer:

(i) Core layer
(ii) Distribution layer
(iii) Access layer

→ Core layer:

- Core layer contain server room and server room have Firewall, Router, 2-L3+ Switches, path Rack, Server inverted etc.

→ Distribution layer:

- Distribution layer contain 4-switches
- The 4 switches are:

(i) Main Build → R7 (Room no - 213, 2nd floor).
(ii) Main Build → R1 (Room no - 301, 3rd Floor).
(iii) Main Build → R6 (Room no - 11B, 1st floor).
(iv) Main Build → R2 (Room no - 201, 2nd floor).

→ This 4- switches contain:

(i) 18 port gigabit Easy Smart Switch with 16- port. [TL-SG1218MPE]

(ii) [SG3428XMP] → 24 port gigabit & 4-bit 10GESFP, L2+ managed switch 24 port PoE+.

→ Now let's discuss these 2 above components:

(i) TP-Link TL-SG1218MPE

* Likely Used For: CCTV Cammera, wifi Access point etc.
* 18 port gigabit "Easy smart" switch.
* 16 port support PoE+.

(ii) TP-link Omada SG3428XMP

* Higher-end managed Layer 2+ Switch
* 24 PoE+ port, Uplinks, 10GSFP
* Usually acts as a main distribution for each floor.

→ Access Layer:

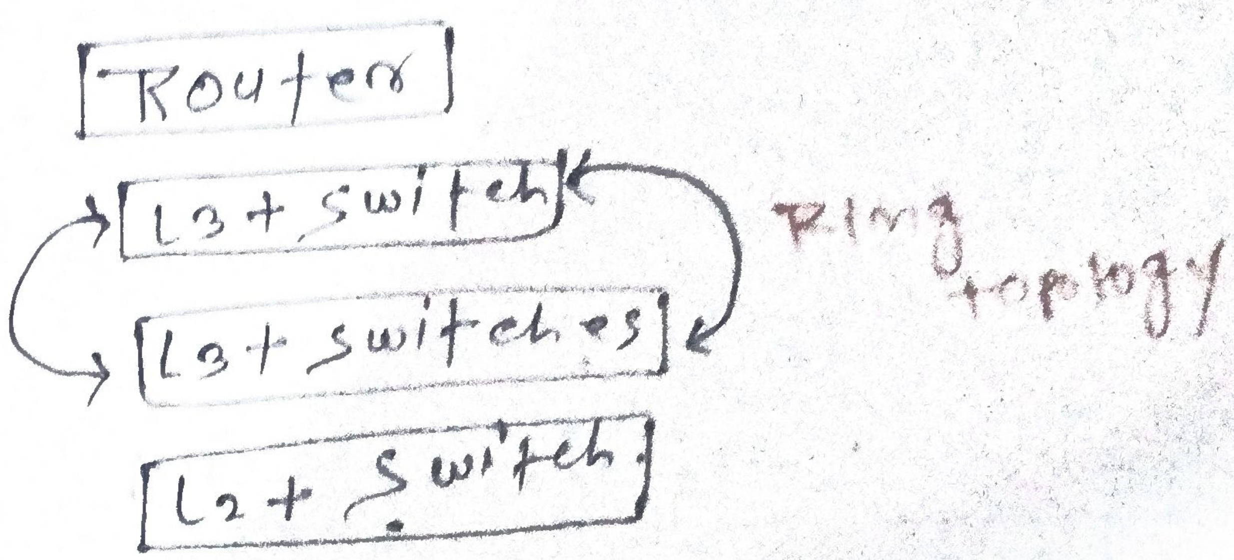
- Access layer contain 3-switches i.e

i) main build R1 - [Room no- 301, 3rd Floor]
ii) main build R4- [Room no- 05, Ground Floor]
iii) main build R5- [Room no- 18, R, Ground floor]

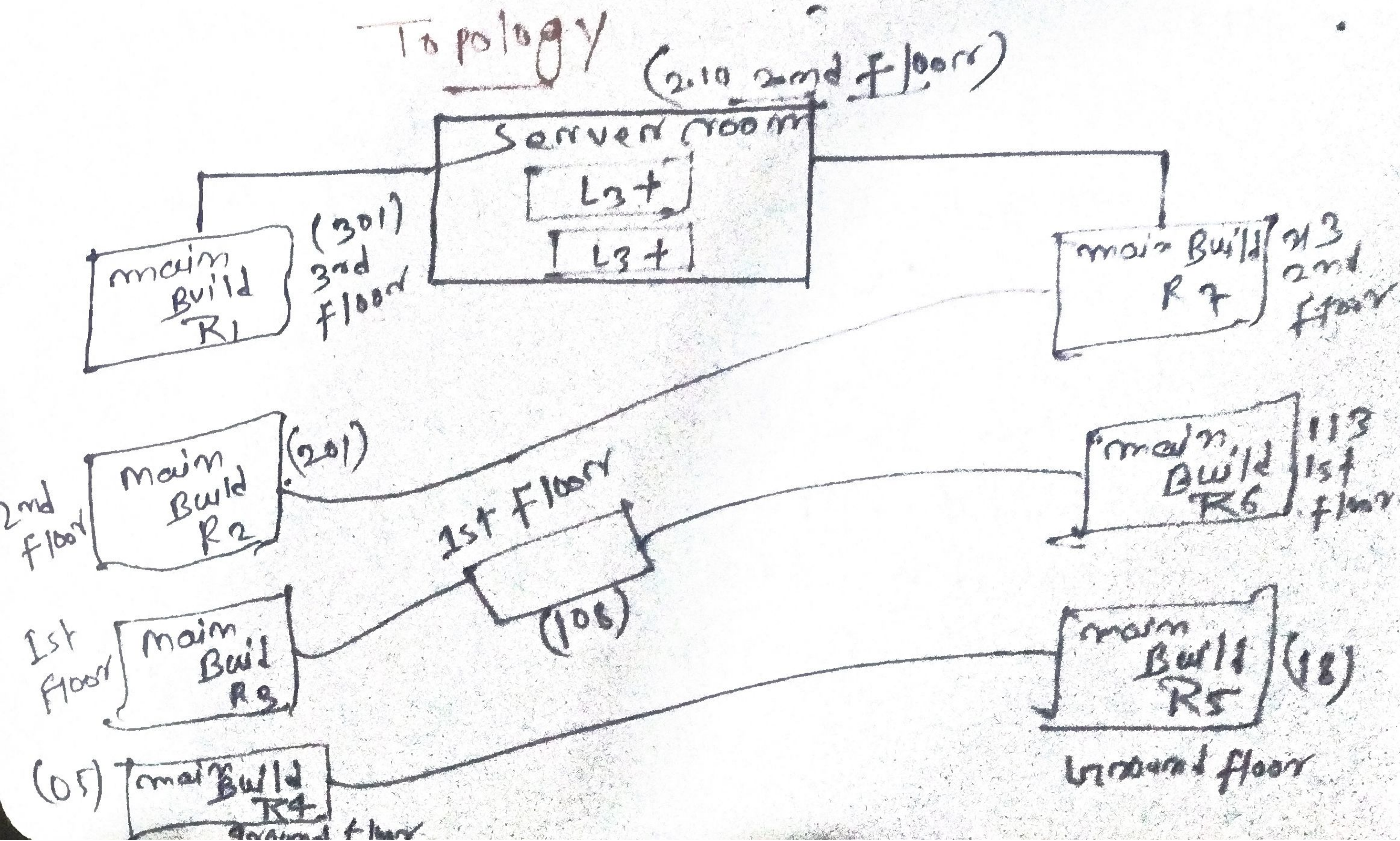
* Every switches act as access layer which helps to provide network connectivity to all classroom, lab, faculty room, etc.

* Access Layer directly connects
i) Ethernet port
ii) wireless Ap
iii) Lab system
iv) outdoor wireless Ap (EX-018)

→ SERVER ROOM Architecture / topology




Topology



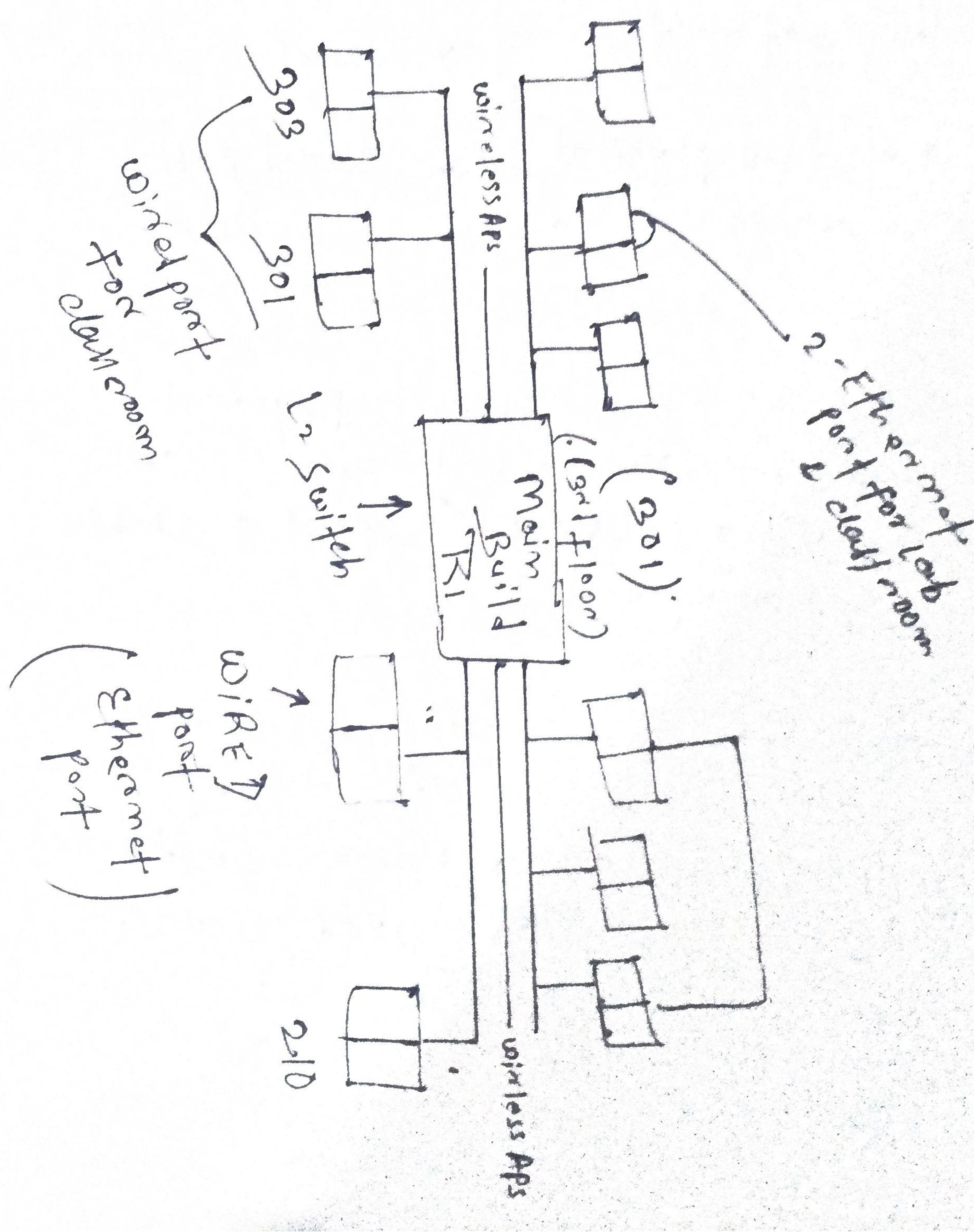
Summary:

→ R1, R7, R2, R6 → Distribution Layer

→ Server room → Core Layer

→ R1, R2, R3, R4, R5, R6, R2 → Access Layer

→ Access Layer Switch (3rd floor, 301)

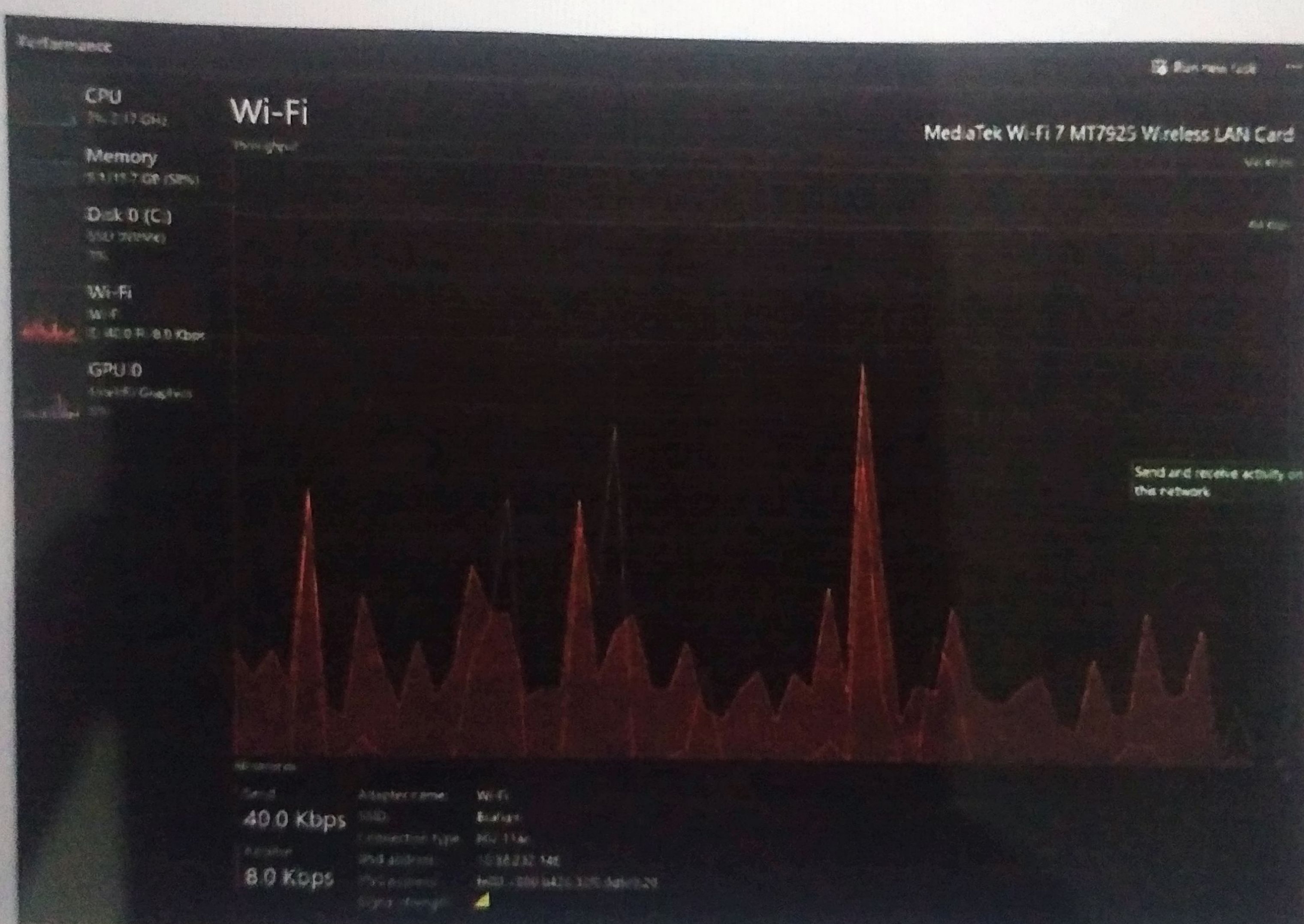


# WIRELESS CONNECTION

```
C:\Users\conne>ipconfig

Windows IP Configuration


Wireless LAN adapter Wi-Fi 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi 3:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::c080:b426:32f0:8dfd%20
   IPv4 Address. . . . . . . . . . . : 10.38.232.146
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.38.232.103

Ethernet adapter Ethernet 3:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2409:408a:1c8b:f98b:b142:f535:a2d7:cf02
   Temporary IPv6 Address. . . . . . : 2409:408a:1c8b:f98b:d76:27d0:d0a4:5185
   Link-local IPv6 Address . . . . . : fe80::3c29:2fe7:1878:986d%9
   IPv4 Address. . . . . . . . . . . : 10.238.72.127
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::d8ed:73ff:fe9f:79de%9
                                       10.238.72.213

C:\Users\conne>
```

↳ using "ipconfig"

WIRELESS SPEED




```
C:\Users\conne>getmac

Physical Address    Transport Name
=================== ==========================================================
04-68-74-96-8D-8F   \Device\Tcpip_{C6528A0B-6572-4CA0-A806-97A8623E6306}
N/A                 Hardware not present
```

→ MAC ADDRESS
→ using "getmac"

# WIRED CONNECTION

```
C:\Users\conne>ipconfig

Windows IP Configuration


Wireless LAN adapter Wi-Fi 2:

   Media State . . . . . . . . . . . :
   Connection-specific DNS Suffix  . : Media disconnected

Wireless LAN adapter Wi-Fi 3:

   Media State . . . . . . . . . . . :
   Connection-specific DNS Suffix  . : Media disconnected

Wireless LAN adapter Wi-Fi:

   Media State . . . . . . . . . . . :
   Connection-specific DNS Suffix  . : Media disconnected

Ethernet adapter Ethernet 3:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . :
   Temporary IPv6 Address. . . . . . : 2409:408a:1c8b:f90b:b142:f535:a2d7:cf02
   Link-local IPv6 Address . . . . . : 2409:408a:1c8b:f90b:d76:2740:d0a4:5186
   IPv4 Address. . . . . . . . . . . : fe80::3c29:2fe7:1870:980d%9
   Subnet Mask . . . . . . . . . . . : 10.238.72.127
   Default Gateway . . . . . . . . . : 255.255.255.0
                                       fe80::d0ed:73ff:fe04:790e%9
                                       10.238.72.213

C:\Users\conne>
```

USING IPCONFIG

→ Difference beetween Wired & Switch wireless speed.

Solⁿ:

| Wired Network | Wireless Network |
| --- | --- |
| * Faster Communication | * Slower Communication |
| * Stable Connection | * affected by distence inteface |
| * low latency | * Higher latency |
| * Dedicated physical medium | * shared Radio medium. |

→ Ethernet provide direct physical Communication Using network Cables resulting in higher speed & stable Performance.

→ Wireless Connection Uses radio wave which are affected by interferance, distance, obstacles & multiple Connected Users.

Q. Role of subnet mask in network ?

Solⁿ:
A subnet mask identifiers the network portion & host porstion of an IP Address. it helps device determine wether another multiple device belong to the same network or different network.

* it is important for:
1. Network segment
2. Efficient routing
3. Communication b/w device.

→ Observation:

During the study of the college network infrastracture:
1. A centralized server room was used of network back bone.
2. Two L3 switch were Connected in redundant cyclic stracture.
3. Managed TP-Link PoE + Switches were Used in network Cabinets

→ Advantage of Existing Network:
* Scalable architecture
* Centralized management
* Supports PoE + device

* Easy expansion of floor & classroom
* Support both wireless & wired
* Redudant link for failourer.

→ Conclusion:

The observed Campus LAN Follow a Hybrid hierarchical topology Consist of tree, Extended Star, Ring. The network uses managed PoE Switches wireless access point Ethernet port Centralized Server infrastrature to provide efficient wired & wireless Communication across the Campus.